

SUMMARY TABLE: CHEMISTRY GRADE 10

Sub-Strand 1.1: Introduction to Chemistry — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.1: Introduction to Chemistry
Driving Question	DRIVING QUESTION: "How does Chemistry explain what happens to the matter around us every day — and why does understanding Chemistry help us make safer, smarter choices in our lives?"

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon & Defining Chemistry: What Happened to the Chapati Dough?	Raw chapati dough placed on a hot jiko transforms irreversibly into a firm, golden, aromatic chapati. In the PhET States of Matter simulation, water particles move from tightly packed (solid) to loosely spaced (gas) and back — showing that some matter changes are reversible at the particle level, while the chapati change is not.	Chemistry is the study of matter, its properties, composition, and the changes it undergoes. Matter is anything that has mass and takes up space. Studying Chemistry helps us understand why and how matter changes — from chapati on a jiko to medicines in our bodies. Some changes in matter are reversible; others (like cooking dough) are not.	Chemistry — and not Biology or Physics alone — is the best tool for explaining the chapati transformation because the change involved the properties, composition, and irreversible chemical change of the matter (dough) itself. Chemistry helps us understand how drugs interact with the body at a molecular level, and why understanding matter changes leads to safer, smarter choices in daily life.
Lesson 2 Branches of Chemistry: Mapping the Science to Our World	Learners observed 7 colour-coded Branch Cards, each describing one branch of Chemistry (Organic, Inorganic, Physical, Analytical, Biochemistry, Nuclear, Industrial) through a specific Kenyan context — including how Bamburi Cement production relates to Inorganic Chemistry, how KEBS food testing relates to Analytical Chemistry, and how the browning of chapati and nyama choma relates to Biochemistry.	Learners learned that Chemistry is divided into specialised branches, each studying a different aspect of matter and its changes. Biochemistry specifically explains the Maillard reaction — the irreversible chemical transformation of proteins and sugars in chapati dough when exposed to heat — which is why a cooked chapati can never be returned to raw dough. Each branch underpins real Kenyan industries, careers, and daily-life events.	Using their expanded concept maps and the Maillard reaction diagram, learners explained that the chapati phenomenon involves Biochemistry because it is a chemical reaction between amino acids (from wheat protein) and sugars (from wheat starch) driven by heat, producing new molecules responsible for the brown colour, aroma, and firm texture. Because new, chemically distinct molecules are formed, the change is irreversible — directly answering part of the driving question: 'Chemistry explains what happens to matter around us every day' through the lens of Biochemistry.

Lesson 3 Chemistry in Agriculture and Food — How Chemistry Feeds Kenya	Learners examined ARES platform resources on fertilisers (NPK/urea) used in Rift Valley maize farming, pesticides (pyrethrum) in Nakuru, soil pH effects on tea in Kericho and maize in Trans Nzoia, and food preservatives (sodium benzoate) and fermentation (lactic acid in uji and mursik). They also observed real Kenyan food product labels (Pembe flour, Bidco oil, Kensalt) identifying preservatives, expiry dates, quality marks, and caution messages.	Chemistry is applied in Kenyan agriculture through fertilisers that change soil chemistry to support crop growth, pesticides that target harmful organisms through chemical reactions, and soil pH management that determines which crops thrive in which regions. In the food industry, Chemistry explains how preservatives prevent spoilage by inhibiting microbial chemical reactions and how fermentation — a chemical change — transforms maize porridge into uji and milk into mursik. These are all examples of Chemistry explaining and controlling changes to matter in daily Kenyan life.	Just as the chapati dough undergoes an irreversible chemical change on the jiko — transforming matter through heat — Chemistry operates in every stage of Kenya's food chain: changing soil to grow the wheat, preserving the flour on the shelf, and fermenting the uji in the pot. Understanding these chemical changes helps Kenyans make informed, safe, and smart choices — as farmers, consumers, and citizens.
Lesson 4 Chemistry in Medicine, Pharmaceuticals, and Manufacturing	Learners read and analysed a structured text on Chemistry in medicine (KNH blood tests, anaesthetics, paracetamol), the pharmaceutical industry (drug synthesis, KEBS standards, quality control), and manufacturing (EABL fermentation, Bidco oil refining). They observed how Chemistry operates as the underlying science in each sector and examined product labels for quality marks, expiry dates, and dosage instructions.	Learners learned that: (1) Chemistry explains how drugs interact with the body at a molecular level, including why incorrect dosage causes harm; (2) the terms drug, prescription, and dosage have precise scientific meanings that protect consumers; (3) KEBS uses chemical quality standards to safeguard Kenyan consumers; (4) manufacturing industries like EABL (fermentation) and Bidco (oil refining) depend on controlled chemical changes — the same type of irreversible change seen when chapati dough is placed on the jiko.	Learners explained how the irreversible chemical change in chapati-making on the jiko is conceptually identical to chemical changes in pharmaceutical manufacturing and in the body — in all cases, new substances are formed at the molecular level and the original matter cannot be recovered. They explained that Chemistry is therefore not only a kitchen science but a life-protecting discipline that underpins medicine, consumer safety, and industry in Kenya.
Lesson 5 Chemistry in Other Industries: Nuclear, Sports, Entertainment & Energy	Learners observed four Kenya-relevant images connecting Chemistry to energy (a nuclear power station; Olkaria geothermal and the KPLC grid), sports nutrition (Kenyan marathon runners using energy gels and electrolytes), entertainment (coloured dyes, metal-salt pyrotechnics at Nairobi festivals), and environmental chemistry (water quality in Lake Victoria, air pollution on Mombasa Road). They observed expert group presentations and populated a spoke diagram connecting all four industries to the central idea of Chemistry as the study of matter and its changes.	Chemistry operates in many industries beyond the laboratory: nuclear energy involves matter changes at the atomic level (fission); sports nutrition uses Chemistry to understand how carbohydrates (like ugali and energy gels) are broken down in the muscles of athletes like Kipchoge; entertainment relies on chemical reactions (synthetic dyes) and heated metal salts, whose excited electrons give out coloured light, to create colour and spectacle; and environmental Chemistry monitors and protects the matter in Kenya's water, soil, and air. In every case, Chemistry is explaining, predicting, or controlling how matter changes — the same process that transformed chapati dough on the jiko.	The central explanation across all industries is that Chemistry is the science of matter change. Whether a uranium atom splits in a reactor, glycogen breaks down in a runner's muscle, a copper compound gives a green flame in a firework, or sulphur dioxide reacts with water vapour in Nairobi's air — all of these are matter undergoing change that Chemistry can describe and explain (the reactor is a nuclear change inside the atom's nucleus; the others are chemical changes). The chemical changes are the same type of change (irreversible, involving new substances and energy transfer) that occurred when raw chapati dough was transformed by heat into a firm golden disc. Understanding these changes allows scientists, engineers, athletes, and

			policy-makers to make safer, smarter decisions — which is exactly what the driving question asks us to explore.
Lesson 6 Careers in Chemistry and Gender Stereotyping	Learners observed (via ARES platform and print research cards) that Chemistry careers span six major sectors — food, pharmaceutical, agricultural, industrial, sports, and nuclear/environmental — and that real Kenyan scientists work in every one of these sectors. They also observed, through their own Phase 1 board tally, that the names they initially associate with 'chemist' are disproportionately male, and that this disparity is a product of social stereotype rather than scientific aptitude.	Learners learned that Chemistry underpins careers that directly affect everyday Kenyan life — from the KEBS food technologist certifying safe unga, to the KEMRI medicinal chemist formulating malaria drugs, to the agrochemist protecting Kericho tea farms. They learned that gender stereotypes in Kenyan society create systemic barriers that reduce equitable access to Chemistry careers, and that self-efficacy — the belief in one's own ability to succeed — is a critical tool for overcoming those barriers. They also consolidated prior learning by connecting each career sector back to the core idea that Chemistry is the study of matter and its changes — including the irreversible change that turns chapati dough into a golden disc.	Learners can now explain, using evidence from their jigsaw research, that Chemistry careers are not defined by gender but by knowledge, skill, and interest — and that the same chemical principles that explain the chapati phenomenon (matter transformation, energy, irreversible change) are applied daily by Kenyan chemists across industries that feed, heal, protect, and empower Kenyan communities.
Lesson 7 Meaning of Drug, Prescription, Dosage and Substance Use	Learners read a 'Science of Drugs' article featuring Kenyan pharmacist Amina Ochieng of Kisumu County Referral Hospital, and analysed printed Panadol and Metformin medicine labels using a structured 'Label Detective' observation sheet, recording active ingredients, dosage, caution statements, expiry dates, and prescription requirements.	Learners defined drug, prescription, dosage, over-the-counter (OTC) medicine, prescription-only medicine, substance use, and substance abuse; identified that Metformin requires a prescription while Panadol does not; and co-constructed a Class Definition Wall of the seven key terms with Kenya-relevant examples.	Learners connected the pharmaceutical branch of Chemistry to the anchoring phenomenon — just as uncontrolled heat causes irreversible changes to chapati dough on the jiko, uncontrolled or excessive intake of chemical substances (drugs) causes irreversible harm to the human body; understanding Chemistry — specifically dosage and prescription information — empowers Kenyans to make safer, smarter choices about their health.
Lesson 8 Effects of Drug and Substance Use: Tobacco — Chemistry of an Irreversible Harm	Learners watched a video showing how tobacco smoke releases tar, nicotine, and carbon monoxide into the body, and annotated a body-system diagram with the specific effects of each chemical on lungs, blood vessels, and the nervous system.	Learners learned that 'natural' does not mean 'safe' — tobacco is a natural plant, yet its chemicals cause specific, largely irreversible damage at the molecular level; tar coats and destroys lung tissue, nicotine creates chemical dependency by acting on nerve receptors, and carbon monoxide binds to haemoglobin and blocks oxygen transport, just as heat irreversibly transforms chapati dough into a fundamentally different substance.	Learners explained, using a Claim–Evidence–Reasoning frame, why the statement 'shisha is natural so it is safe' is chemically incorrect, connecting evidence from the video to the anchoring phenomenon: Chemistry governs all matter — living and non-living — and the same principle that explains why chapati dough cannot return to its original form explains why tobacco damage to lung tissue and haemoglobin cannot be easily undone.

Lesson 9 Consumer Protection and the Safe Learning Environment	Learners examined labels on real Kenyan manufactured products — including Jogoo maize flour, Jik bleach, and paracetamol packaging — recording KEBS quality marks, expiry dates, ingredient lists, E-number preservative codes, nutritional panels, dosage instructions, and safety/caution warnings on a structured Label Analysis Recording Sheet.	Learners learned that Chemistry is embedded in every manufactured product through quality standards, preservative chemistry, dosage pharmacology, and hazard communication; that consumer protection rights (to accurate information, quality assurance, safety warnings) are grounded in Chemistry knowledge; and that reading and interpreting chemical labels is itself a Chemistry skill that enables safer, smarter daily choices.	Learners explained how Chemistry knowledge creates both rights (e.g., the right to know what preservatives are in food, the right to accurate dosage information) and responsibilities (e.g., the responsibility to read safety warnings, to avoid misusing lab or household chemicals, to follow correct dosage), and connected the chapati phenomenon's flour-to-product journey to the full chain of consumer protection Chemistry — from farm to factory to label to jiko to body.
Lesson 10 Project: Poster on Drug & Substance Use — Unit Synthesis & Final Explanation	Learners observed (Lesson 1): raw chapati dough placed on a hot jiko transforms irreversibly into a firm, golden, aromatic disc. The original soft dough cannot be recovered. This simple, familiar Kenyan kitchen event launched ten lessons of inquiry.	Learners learned: Chemistry is the scientific study of matter and its changes — physical and chemical. The branches of Chemistry (organic, inorganic, physical, biochemistry, analytical, nuclear, and others) explain different aspects of how matter behaves. Chemistry underpins agriculture in the Rift Valley, pharmaceutical production, food processing (githeri, ugali, mandazi), manufacturing, medicine, and sport. Drugs are chemical substances that interact with body matter at a molecular level; prescription, dosage, and responsible use are Chemistry-informed concepts that protect health. Matter exists as solid, liquid, or gas, with particle arrangement determining properties. Understanding Chemistry empowers learners to make safer, smarter choices in their lives.	The driving question is answered: Chemistry explains what happens to the matter around us every day by studying the composition, properties, and changes of matter — from the irreversible chemical transformation of chapati dough on a jiko, to the way pharmaceutical drugs interact with body cells. Understanding Chemistry helps us make safer, smarter choices because it gives us evidence-based tools to evaluate food labels, understand drug prescriptions and dosages, recognise the risks of substance misuse, and appreciate the role of Chemistry-based industries in our daily Kenyan lives. The chapati cannot become dough again because a chemical change has occurred — bonds have broken and new substances have formed. Chemistry is the science that explains this, and every branch of Chemistry adds another layer of understanding to the matter around us.

ATTRIBUTION AND LICENSING

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